Project Scenario: CloudNova Enterprise AWS Infrastructure

CloudNova Solutions is a growing technology company that hosts business applications and manages sensitive customer data on AWS. As the company expands, its engineering, data, and operations teams need a secure and reliable cloud environment.

Currently, the company has separate environments for its application servers, internal data services, and IT operations. These environments need to communicate securely, share files, store backups, and respond automatically when infrastructure problems occur.

The company has asked your team, as its AWS Cloud Engineers, to design and implement a cloud infrastructure that supports these business needs.

The application team needs to run a web application on private EC2 instances while allowing customers to access it through a public entry point. The data team needs to maintain isolated resources but still allow authorized communication with the application environment. The operations team needs centralized visibility into infrastructure health and a secure way to administer resources.

CloudNova also needs a reliable way for application servers to share files, store backups without sending S3 traffic through a NAT Gateway, and notify the operations team when important infrastructure issues arise. For selected failures, the company wants an automated response rather than relying entirely on manual intervention.

As the number of environments grows, management is also concerned about how the network will scale, how access between environments will be controlled, and how to keep infrastructure costs under control.

Your team must develop an AWS solution that addresses these business challenges using appropriate networking, storage, monitoring, notification, and automation services. You must justify your architectural decisions and demonstrate how the resulting environment behaves during normal operation and when failures occur.

Your mission: Build a secure, scalable, monitored, and partially self-healing AWS environment for CloudNova Solutions.